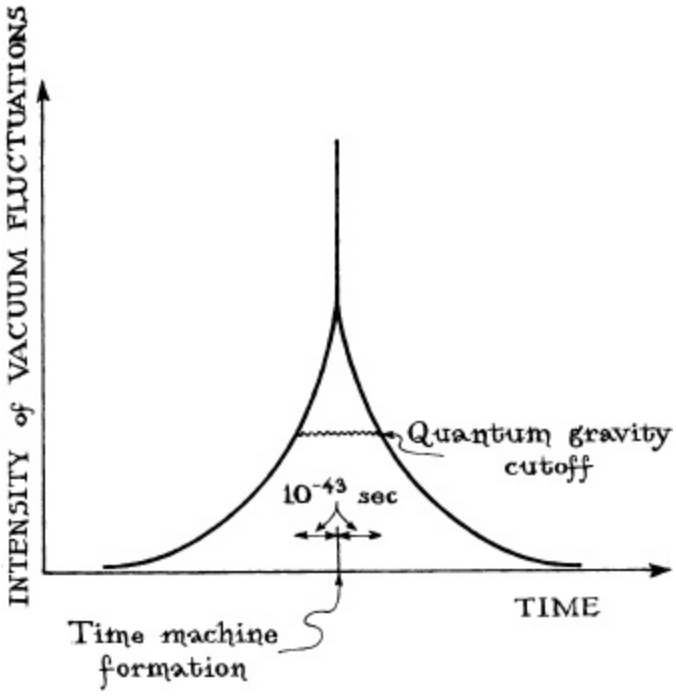


14.12 Evolution of the intensity of the electromagnetic vacuum fluctuations that circulate through a wormhole just before and just after the wormhole becomes a time machine.

In September 1990, when we submitted our manuscript to the *Physical Review,* Kim and I sent copies to a number of colleagues, including Stephen Hawking. Hawking read our manuscript with interest—and disagreed. Hawking had no quarrel with our calculation of the beam of circulating vacuum fluctuations (and, in fact, a similar calculation by Valery Frolov in Moscow had by then verified our results). Hawking's quarrel was with our analysis of quantum gravity's effects.

Hawking agreed that quantum gravity was likely to cut off the growth of the vacuum fluctuations 10^{-43} second before the time machine is created, that is, 10^{-43} second before they otherwise would become infinitely strong. "But 10^{-43} second as measured by whom? In whose reference frame?" he asked. Time is "relative," not absolute, Hawking reminded us; it depends on one's reference frame. Kim and I had assumed that the appropriate reference frame was that of somebody at rest in the wormhole throat. Hawking argued, instead (in effect), for a different choice of reference frame: that of the fluctuations themselves—or, stated more precisely, the reference frame of an observer who circulates, along with the fluctuations,